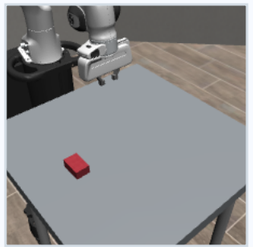
DB0 v0.4 · Active-Full-13-v0.3 · representative frozen environment views

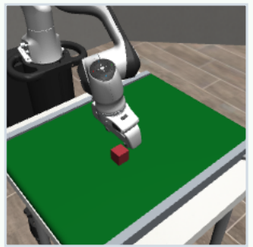
P0 Future Pose



P0

robot still
8 future SE(3)
50–400 ms
forecast only

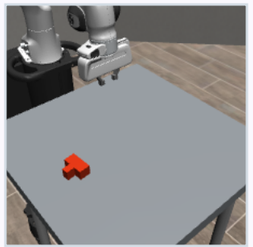
T1 Belt Capture



T1-Belt

intercept
grasp + lift
stable hold
moving support

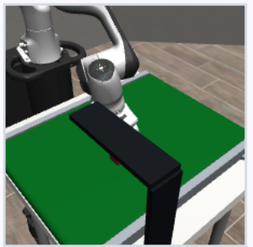
T1 SE(3) Split



T1-XYZ / SO3

matched XYZ
independent yaw
causal contrast
T-shaped target

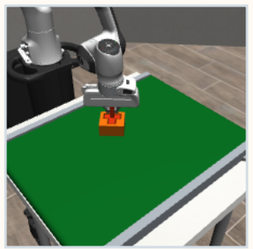
T1 Occlusion



T1-OCC

physical screen
hidden interval
history needed
no pose oracle

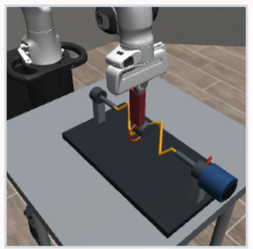
T2 Moving Goal



T2-TRANS / SE3

future release
containment
settle + dwell
moving receptacle

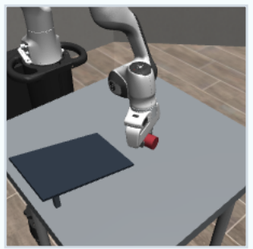
T3 Buzz Wire



T3-Phase / Full

phase align
traverse loop
no wire touch
rotating obstacle

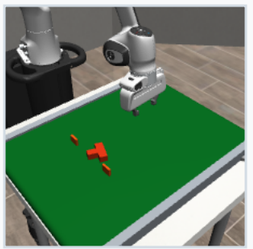
T4 Free Physics



T4-Sphere
Slider / Can

ramp exit
unsupported catch
roll / slide / slip

T5 Impact Replan



T5-Replan

branch impact
observe + correct
keyed grasp
post-event state

Shared physical clock
20 Hz RGB-D/control · 500 Hz physics · waiting, inference, actuation, contact, and dwell all consume time

Separate reports
forecast · SR · TC · CT · safety · latency